

SALES PERFORMANCE ANALYSIS AND INTERACTIVE DASHBOARD DEVELOPMENT

A Data Analytics Portfolio Project

Prepared by

Esther Oluwapelumi Oyeyemi

Junior Data Analyst

Tools Used

- Microsoft Excel
- Power BI

Dataset

Sample Superstore Data

Date

(July 2026)

Table of Contents

Contents

1.0 Executive Summary.....	3
2.0 Introduction.....	3
2.1 Business Problem	4
2.2 Project Objectives	4
3.0 Dataset Overview	5
3.1 Dataset Description	5
3.2 Dataset Source.....	5
3.3 Dataset Summary	6
3.4 Description of Variables	6
3.5 Why This Dataset Was Chosen.....	6
4.0 Data Cleaning and Preparation	7
4.1 Overview.....	7
4.2 Data Cleaning Process	7
4.3 Missing Values	7
4.4 Duplicate Records.....	7
4.5 Data Types	8
4.6 Data Preparation in Power BI.....	8
4.7 DAX Measures	8
4.8 Dashboard Development	9
4.9: Key Findings.....	14
5.0 RECOMMENDATIONS	15
5.1 Increase Inventory Before Peak Sales Period	15
5.2 Focus on High-Performing Cities.....	15
5.3 Review the Pricing and Discount Strategy for Tables.....	15
5.4 Improve the Profitability of Bookcases	15
5.5 Promote Frequently Purchased Products.....	16
5.6 Maintain Efficient Standard Class Shipping.....	16
6.1 Conclusion	16
References.....	17

Executive Summary

1.0 Executive Summary

For this project, I analyzed sales data from the **Sample Superstore** dataset to understand how the business was performing and to identify areas where performance could be improved. The goal was to turn raw sales data into meaningful insights that could help management make better business decisions.

The dataset contains **9,994 sales records** and **21 columns**, including customer details, product information, sales, profit, discounts, regions and order dates. Before starting the analysis, I reviewed the dataset to ensure the data was complete and in the correct format for analysis.

I used **Microsoft Power BI** to build an interactive dashboard. The dashboard includes key performance indicators (KPIs), charts, maps and filters that allow users to explore sales performance from different perspectives. It covers areas such as overall sales, product performance, regional performance, sales trends, profitability, and business recommendations.

From the analysis, I was able to identify the products and regions that performed best, as well as products that were generating losses. I also observed sales patterns over time and highlighted opportunities that could help improve future business performance.

Overall, this project allowed me to apply the skills I have learned in **Excel** and **Power BI** to solve a real business problem. It also strengthened my understanding of data analysis, data visualization, and how data can be used to support better business decisions.

2.0 Introduction

Sales data contains valuable information that can help businesses understand their performance and make better decisions. However, when sales records are stored as raw data, it can be difficult to identify trends, compare performance or spot areas that need attention.

In this project, I analyzed the **Sample Superstore** dataset to better understand the company's sales performance. I analyzed the dataset using **Microsoft Excel** for data preparation and **Power BI** was used to build an interactive dashboard that presents the results in a clear format.

The dashboard allows users to monitor important business metrics such as revenue, profit, product performance, regional sales and sales trends even if they are non-tech users. It also

highlights areas where the business is performing well and areas that may require attention and improvement. The goal was not only to present the data but also to provide insights that could support better business decision making.

2.1 Business Problem

Although the company collects sales data from its daily operations, the information is stored as individual transaction records. This makes it difficult to quickly understand overall business performance or identify important trends without spending time analyzing the data.

Management needs a simple way to answer questions such as:

- Which products generate the highest revenue?
- Which products are making a loss?
- Which regions perform best?
- How have sales changed over time?
- Where should the business focus its efforts to improve performance?

Without a centralized dashboard, answering these questions can be slow and time-consuming. This project addresses that challenge by transforming raw sales data into an interactive dashboard that provides clear insights and supports faster data-driven decision-making.

2.2 Project Objectives

The main goal of this project was to analyze the company's sales data and create an interactive dashboard that makes it easier to understand business performance.

The specific objectives were to:

- Analyze the overall sales performance of the company.
- Calculate key performance indicators (KPIs) such as Total Revenue, Total Profit, Total Orders and Average Revenue per Sale.
- Identify the best-performing products and categories.
- Compare sales performance across different regions, states and cities.
- Examine sales trends over time.
- Identify products and categories with low or negative profitability.

- Present the findings through an interactive Power BI dashboard.
- Provide practical recommendations based on the analysis.

By achieving these objectives, the project aims to provide a clearer understanding of the company's sales performance and support better business decisions based on data rather than assumptions.

3.0 Dataset Overview

3.1 Dataset Description

For this project, I used the **Sample Superstore** dataset, a retail sales dataset that contains information about customer orders, products, sales, shipping and profitability. The dataset is widely used for data analysis and visualization projects because it represents real-world business transactions and allows analysts like myself to answer varieties of business questions.

The dataset contains **9,994 records** and **21 columns**. Each row represents a single customer order and includes details such as the order date, customer information, product details, sales amount, quantity purchased, discount given and profit earned.

The dataset provides enough information to evaluate the company's sales performance, identify profitable and loss-making products, compare regional performance, and analyze sales trends over time.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gut Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10	Furniture	Bookcases	
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gut Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10	Furniture	Chairs	
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Var Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10	Office Supplies	Labels	
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donoghue Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10	Furniture	Tables	
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donoghue Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10	Office Supplies	Storage	
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hernandez Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10	Furniture	Furnishings	
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hernandez Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10	Office Supplies	Art	
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hernandez Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10	Technology	Phones	
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hernandez Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10	Office Supplies	Binders	
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hernandez Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10	Office Supplies	Appliance	
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hernandez Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10	Furniture	Tables	
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hernandez Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10	Technology	Phones	
13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew A Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10	Office Supplies	Paper	

Figure 3.1: Sample Superstore Dataset (First Few Records)

3.2 Dataset Source

The dataset used for this project is the **Sample Superstore** dataset from Kaggle, which is commonly used for learning and practicing data analysis and business intelligence.

3.3 Dataset Summary

Items	Details
Dataset Name	Sample Superstore
Number of Records	9994
Number of Columns	21
Industry	Retail
Data Type	Structured
Tools Used	Excel, SQL, Power BI
Purpose	Sales Performance Analysis

3.4 Description of Variables

Column	Description
Row ID	Unique number assigned to each record
Order ID	Unique identifier for each order
Order Date	Date the order was placed
Ship Date	Date the order was shipped
Ship Mode	Shipping method used
Customer ID	Unique customer identifier
Customer Name	Name of the customer
Segment	Customer segment
Country	Country where the order was placed
City	Customer's city
State	Customer's state
Postal Code	Postal code
Region	Sales region
Product ID	Unique product identifier
Category	Product category
Sub-Category	Product sub-category
Product Name	Name of the product
Sales	Revenue generated from the order
Quantity	Number of items sold
Discount	Discount applied
Profit	Profit earned from the sale

3.5 Why This Dataset Was Chosen

I selected the Sample Superstore dataset because it contains a wide range of sales-related information that makes it suitable for business analysis. It includes customer details, product information, sales, profit, discounts and geographical data, making it possible to explore different aspects of business performance. It also provides enough data to practice SQL

queries, build interactive dashboards in Power BI, and generate meaningful business insights.

4.0 Data Cleaning and Preparation

4.1 Overview

Before starting the analysis, I examined the dataset to make sure it was suitable for reporting and visualization. This involved checking the quality of the data, confirming that the columns had the correct data types and ensuring that the dataset was consistent. The cleaning process helped improve the accuracy of the analysis and reduced the chances of errors during dashboard development.

4.2 Data Cleaning Process

Task	Purpose
Checked for missing values	To ensure all required information was available for analysis
Checked for duplicate records	To confirm that each transaction was unique.
Verified data types	To make sure dates, numbers, and text fields were correctly formatted.
Reviewed column names	To ensure column names were clear and consistent.
Validated numerical fields	To confirm that Sales, Profit, Quantity, and Discount contained valid values.
Imported the cleaned dataset into Power BI	To prepare the data for modeling and visualization.

4.3 Missing Values

I checked the dataset for missing values before beginning the analysis. The review showed that the dataset did not contain any missing values that would affect the results of the analysis. This meant that no imputation or replacement of missing data was required.

4.4 Duplicate Records

I also checked the dataset for duplicate records to ensure that each transaction appeared only once. No duplicate records were found, so no rows needed to be removed.

4.5 Data Types

Column	Data Type
Order Date	Date
Ship Date	Date
Sales	Fixed Decimal Number
Profit	Fixed Decimal Number
Quantity	Whole number
Discount	Fixed Decimal Number
Customer Name	Text
Product Name	Text
Region	Text
Category	Text

4.6 Data Preparation in Power BI

After confirming that the dataset was clean, I imported it into Microsoft Power BI. During the import process, I reviewed the data model, confirmed that the data types remained correct and prepared the dataset for analysis by creating the measures required for the dashboard.

4.7 DAX Measures

Measure	Purpose
Total Revenue	Calculates total sales revenue
Total Profit	Calculates total profit
Total Orders	Counts the total number of orders
Average Revenue	Calculates the average sales value
Average Revenue per Sale	Calculates the average revenue generated per transaction
Profit Margin	Calculates the percentage of profit earned from total sales.

I also created a data table in order to give the date a clean Year/Month sorting.

Total Revenue = SUM(orders[Sales])

Total Profit = SUM(orders[Profit])

Total Orders = DISTINCTCOUNT(orders[Order ID])

Average Revenue per Sale =DIVIDE([Total Revenue],COUNTROWS[Orders],0)

Profit Margin (%) =DIVIDE([Total Profit],[Total Sales], 0)

These measures were used to create KPI cards and other dashboard visuals, allowing users to quickly monitor the company's overall performance and identify important trends.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State
43	CA-2016-101343	Sunday, July 17, 2016	Friday, July 22, 2016	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	California
514	CA-2017-163405	Thursday, December 21, 2017	Monday, December 25, 2017	Standard Class	BN-11515	Bradley Nguyen	Consumer	United States	Los Angeles	California
515	CA-2017-163405	Thursday, December 21, 2017	Monday, December 25, 2017	Standard Class	BN-11515	Bradley Nguyen	Consumer	United States	Los Angeles	California
1606	US-2016-115819	Tuesday, April 19, 2016	Sunday, April 24, 2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California
1607	US-2016-115819	Tuesday, April 19, 2016	Sunday, April 24, 2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California
1609	US-2016-115819	Tuesday, April 19, 2016	Sunday, April 24, 2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California
1807	US-2016-116729	Sunday, December 25, 2016	Wednesday, December 28, 2016	First Class	GK-14620	Grace Kelly	Corporate	United States	Los Angeles	California
2060	CA-2014-106439	Friday, October 31, 2014	Tuesday, November 4, 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California
2061	CA-2014-106439	Friday, October 31, 2014	Tuesday, November 4, 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California
2063	CA-2014-106439	Friday, October 31, 2014	Tuesday, November 4, 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California
2065	CA-2014-106439	Friday, October 31, 2014	Tuesday, November 4, 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California
2068	CA-2014-106439	Friday, October 31, 2014	Tuesday, November 4, 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California
2191	CA-2016-118913	Saturday, June 25, 2016	Wednesday, June 29, 2016	Standard Class	AS-10240	Alan Shonely	Consumer	United States	Los Angeles	California
2492	CA-2014-113579	Saturday, December 13, 2014	Monday, December 15, 2014	Second Class	KD-16345	Katherine Ducich	Consumer	United States	Los Angeles	California
2944	CA-2017-126242	Sunday, November 19, 2017	Friday, November 24, 2017	Standard Class	MC-18100	Mick Crebagga	Consumer	United States	Los Angeles	California
3534	CA-2014-110849	Friday, April 18, 2014	Wednesday, April 23, 2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California
3535	CA-2014-110849	Friday, April 18, 2014	Wednesday, April 23, 2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California
3536	CA-2014-110849	Friday, April 18, 2014	Wednesday, April 23, 2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California
3565	CA-2016-130029	Sunday, July 3, 2016	Wednesday, July 6, 2016	First Class	GT-14755	Guy Thornton	Consumer	United States	Los Angeles	California
3566	CA-2016-130029	Sunday, July 3, 2016	Wednesday, July 6, 2016	First Class	GT-14755	Guy Thornton	Consumer	United States	Los Angeles	California
3759	CA-2015-167745	Friday, September 18, 2015	Wednesday, September 23, 2015	Standard Class	GB-14530	George Bell	Corporate	United States	Los Angeles	California
3760	CA-2015-167745	Friday, September 18, 2015	Wednesday, September 23, 2015	Standard Class	GB-14530	George Bell	Corporate	United States	Los Angeles	California
3844	CA-2014-101931	Tuesday, October 28, 2014	Friday, October 31, 2014	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California
3845	CA-2014-101931	Tuesday, October 28, 2014	Friday, October 31, 2014	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California
3847	CA-2014-101931	Tuesday, October 28, 2014	Friday, October 31, 2014	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California
4339	CA-2015-125234	Friday, November 27, 2015	Tuesday, December 1, 2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Los Angeles	California
4481	CA-2016-109365	Thursday, November 3, 2016	Tuesday, November 8, 2016	Standard Class	XP-21865	Xylona Preis	Consumer	United States	Los Angeles	California

Figure 4.1: Dataset in Power BI

4.8 Dashboard Development

The dashboard was designed to provide a clear and interactive overview of the company's sales performance. It brings together key business metrics and visualizations that allow users to monitor performance, identify trends and make informed business decisions.

Several visuals were used to answer the business questions identified at the beginning of this project. Each visual was carefully selected based on the type of information being presented.

The dashboard includes the following visuals:

Visual	Purpose
KPI Cards	Display Total Sales, Total Profit, Total Orders, Average Sales and Profit Margin.
Line Chart	Shows sales trends over time, making it easy to identify.
Bar Chart	Compares sales or profit across different product categories and sub-categories.
Donut Chart	Shows the contribution of each customer segment or product category to total sales.

Table Visual	Provides detailed information for further analysis.
Slicers	Allows users to filter the dashboard by Region, Category, Segment, State, and other dimensions.

The dashboard was designed with a consistent color theme and a simple layout to make it easy to read and navigate. Interactive filters were also added so anyone looking at the visual can explore the data from different perspectives without creating additional reports.

The Business Overview dashboard provides management with a high-level summary of the company's performance by displaying revenue, profit, order volume, quantity sold, regional contribution and interactive filters.

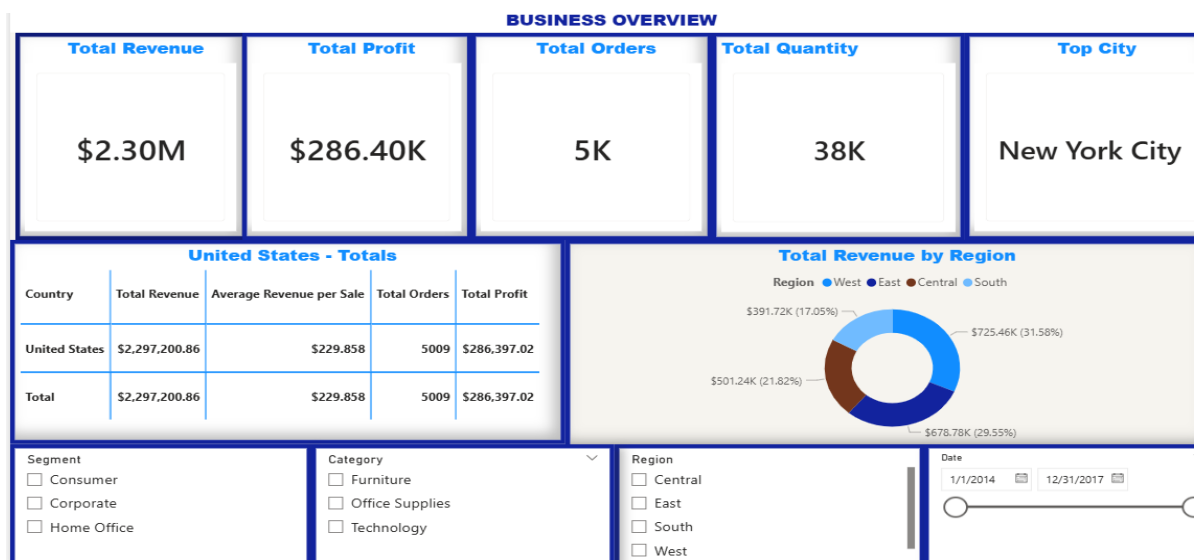


Figure 4.2: Business Overview

The Product dashboard shows performance of each product and product sales

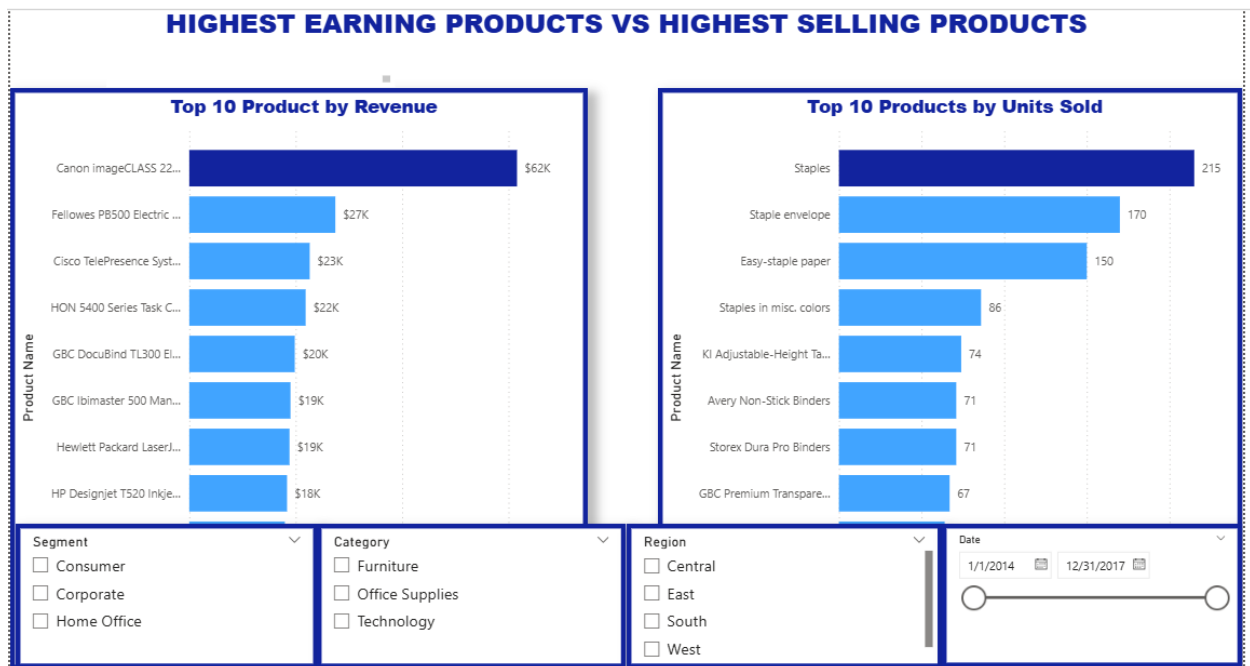


Figure 4.3: Product

The Category dashboard shows the performance of each category.

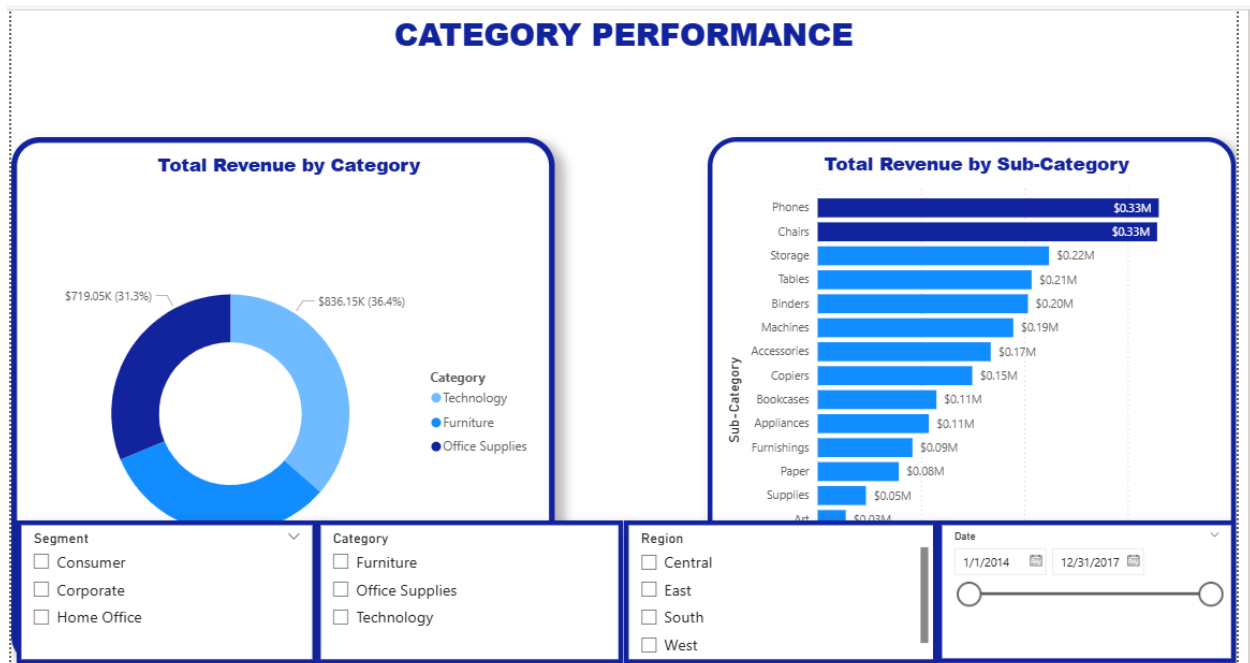


Figure 4.4: Category

REVENUE BY GEOGRAPHY

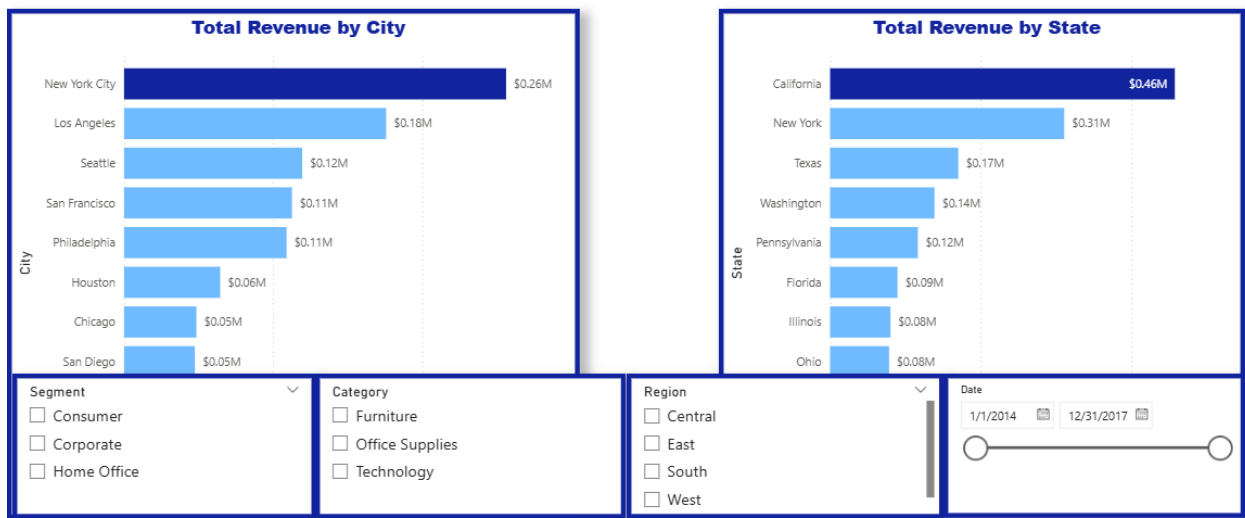


Figure 4.5: Geographic Analysis

REVENUE OVER TIME

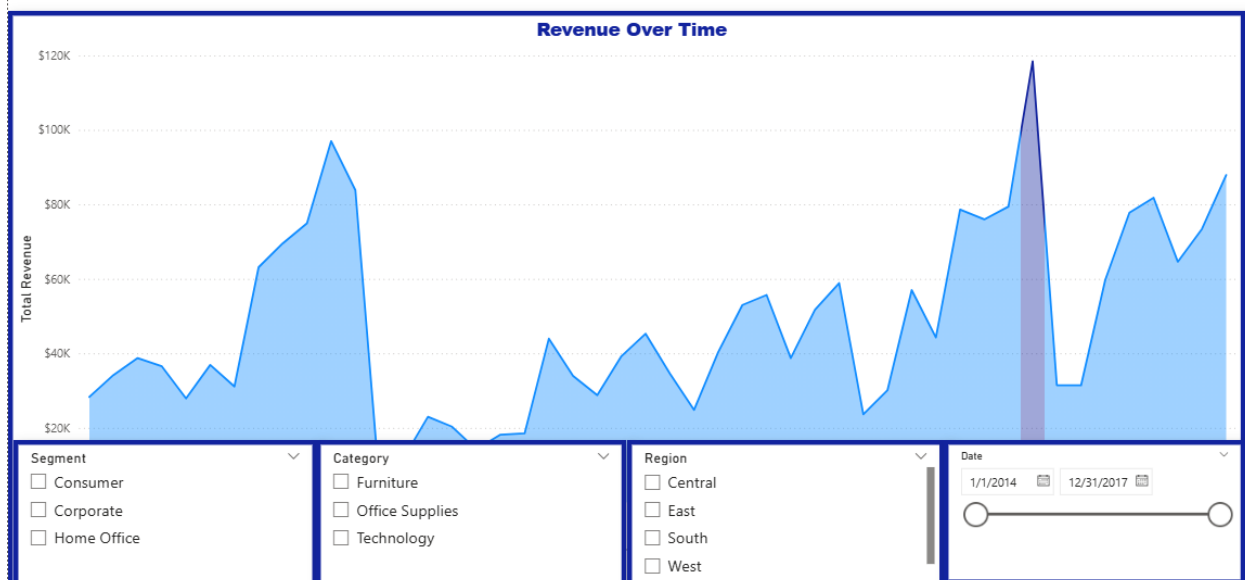


Figure 4.6: Revenue Trend

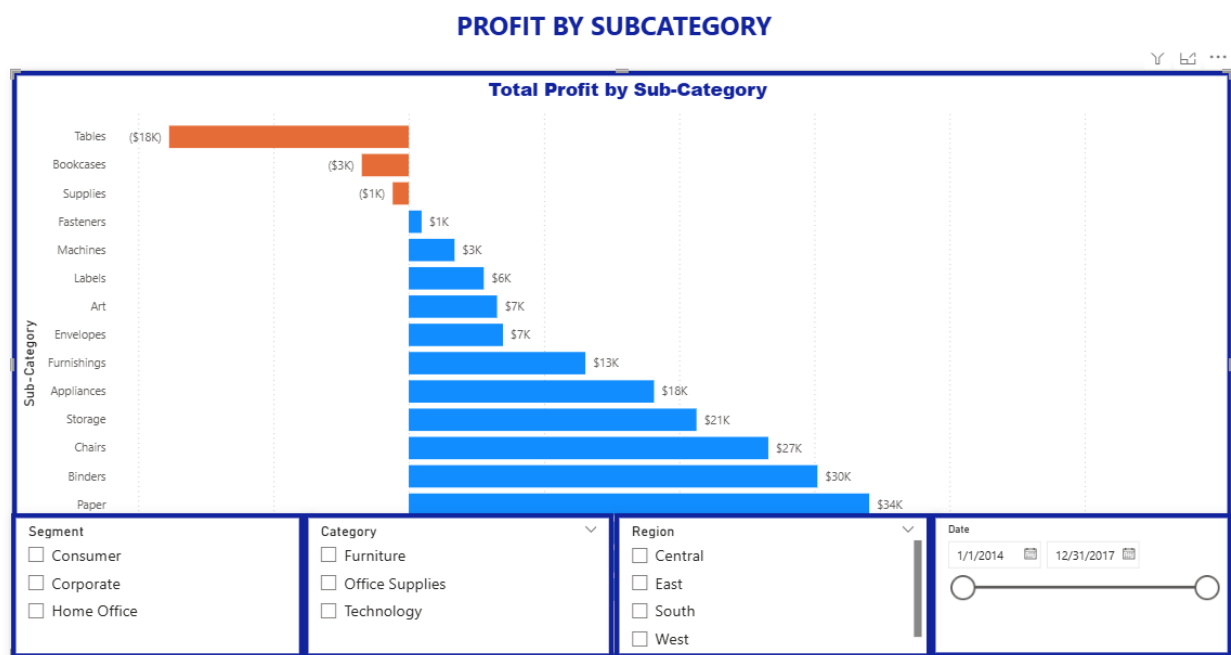


Figure 4.7: Profit Analysis



Figure 4.8: Business Recommendations

4.9: Key Findings

The analysis of the Superstore sales data revealed several important insights into the company's sales performance, customer purchasing behavior, product profitability and regional performance.

Revenue Performance

- The company generated a total revenue of **\$2.30 million** from **5,009 orders** sold between **2014 and 2017**.
- A total profit of **\$286.40K** was recorded from over **38,000 units** sold.

Regional Performance

- The **West Region** generated the highest revenue, contributing about **31.58%** of total sales.
- The **South Region** recorded the lowest revenue among all regions with **17.05%** of total sales.

Geographic Performance

- **New York City** generated the highest revenue among all cities.
- At the state level, **California** was the highest revenue-generating state.

Category Performance

- **Technology** generated the highest revenue among all product categories, indicating strong customer demand for technology products.
- **Office Supplies** contributed steady sales across the business.
- **Furniture** produced strong sales but lower profitability due to losses in some sub-categories.

Product Performance

- The **Canon imageCLASS printer** generated the highest revenue among all products.
- **Staples** was the most frequently purchased product by quantity.

Profitability

- Although sales were high across many products, not every product generated profit.
- **Tables** recorded the largest overall loss.
- **Bookcases** also generated negative profit despite reasonable sales.

Sales Trend

- Revenue generally increased over the years.
- Sales consistently peaked between **September and November**, suggesting a seasonal increase in customer demand during this period.

5.0 RECOMMENDATIONS

The recommendations below are based on the findings from the sales analysis and are aimed at improving the company's sales performance and profitability.

5.1 Increase Inventory Before Peak Sales Period

The analysis showed that revenue consistently increased between September and November, with the highest sales recorded in November. To avoid stock shortages during this period, the company should increase inventory levels of its best-selling products and ensure that warehouse and delivery teams are adequately prepared to meet customer demand.

5.2 Focus on High-Performing Cities

New York City, Los Angeles, and Seattle generated the highest revenue among all cities analyzed. The company should strengthen its presence in these locations through targeted marketing campaigns, customer loyalty programs and improved delivery services to retain existing customers and encourage repeat purchases.

5.3 Review the Pricing and Discount Strategy for Tables

Although the Tables sub-category generated considerable sales, it recorded the highest overall loss. This suggests that high discounts or production costs may be reducing profitability. The company should review its pricing strategy, negotiate better supplier costs or reduce unnecessary discounts to improve profit margins.

5.4 Improve the Profitability of Bookcases

Bookcases also recorded a negative profit despite generating sales. Management should review the pricing structure, supplier costs and promotional strategies for this product line. Packaging bookcases with profitable products or limiting excessive discounts may help improve overall profitability.

5.5 Promote Frequently Purchased Products

Products such as Staples, Staple Envelopes, and Paper were purchased in large quantities. These products can be used in promotional bundles with higher-value items to increase the average order value while maintaining customer satisfaction.

5.6 Maintain Efficient Standard Class Shipping

The analysis showed that Standard Class is the most commonly used shipping method. The company should continue investing in the efficiency and reliability of this shipping option while offering premium shipping services as optional upgrades for customers who require faster delivery.

6.1 Conclusion

This project analyzed the Superstore sales dataset to evaluate the company's sales performance, profitability, customer purchasing patterns, product performance, and regional trends. The dataset was cleaned and transformed using Microsoft Excel and Power Query before being visualized in Power BI through interactive dashboards.

The analysis showed that the business generated over **\$2.29 million** in revenue and **\$286.40 thousand** in profit from **5,009** orders between **2014 and 2017**. The findings revealed that the **West Region and New York City** were the strongest revenue contributors, while the **Technology** category generated the highest sales among all product categories.

The project also identified areas that require management attention. Although some products generated high sales, sub-categories such as **Tables, Bookcases and Supplies** recorded losses, indicating the need to review pricing and discount strategies. In addition, the analysis showed a clear increase in sales between **September** and **November**, providing an opportunity for better inventory planning and sales forecasting.

Overall, this project demonstrates how data analytics can be used to transform raw business data into meaningful insights that support informed decision-making. By applying data cleaning, data modeling, DAX calculations, and interactive dashboard design in Power BI, the project provides valuable recommendations that can help improve sales performance, increase profitability and support future business growth.

References

- Kaggle. Sample Superstore Dataset.
- Microsoft. (2024). Power BI Documentation.
- Microsoft. (2024). Power Query Documentation.
- Microsoft. (2024). Data Analysis Expressions (DAX) Documentation.